

Victoria Knapp Pérez

EDUCATION

2021-PRESENT	Ph.D. Physics, University of California, Irvine, USA (Expected: June 2026) GPA: 4.00/4.00. Advisor: Prof. Timothy M.P. Tait
2021-2023	M.S. in Physics, University of California, Irvine, USA GPA: 4.00/4.00.
2015-2020	B.Sc. in Physics, Universidad Nacional Autónoma de Mexico (UNAM), México GPA: 9.88/10. Thesis Title: "The QCD Critical End Point for Partially Thermalized Systems." Cum Laude. Advisor: Prof. Alejandro Ayala.

SHORT-TERM RESEARCH POSITIONS

JUN-JUL 2019	Summer Research Intern, International Center for Theoretical Physics (ICTP), Trieste, Italy Studied the allowed parameter space of the QCD axion decay constant and the Hubble parameter during inflation using CMB isocurvature perturbation constraints; supervised by Profs. Giovanni Villadoro and Takeshi Kobayashi.
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AWARDS AND SCHOLARSHIPS

MAY 2025	PITT PACC Travel Award Travel allowance for the Phenomenology Symposium 2025, University of Pittsburgh.
JAN 2025	UCI Miguel Velez Scholarship Awarded annually to Latin American graduate students who demonstrate outstanding past academic achievement as well as future promise, have financial need, and are citizens of a Latin American country.
JUL 2024	UC President's Lindau Nobel Laureate Meetings Fellowship Funded participation at the Lindau Nobel Laureate Meetings; interactions with Nobel Laureates and international scientists.
APR 2024	UCI Latino Excellence and Achievement Award (LEAD) Awarded annually to one Physical Sciences graduate student for leadership and support of Hispanic and Latinx student success and research excellence at UCI.
JUN 2022	UCI Department of Physics and Astronomy Outstanding Graduate Student Award Awarded annually to one first-year graduate student in Physics and Astronomy for outstanding academic achievement.
SEP 2021	UCI Chancellor's Fellowship Fellowship for incoming graduate students.
MAY-DEC 2019	UNAM DGAPA-Papiit Scholarship Undergraduate thesis scholarship under supervision of a UNAM researcher.
DEC 2018	"Estancias de Verano teóricas 2019" Contest Fully-funded research stay at ICTP in summer 2019.
2013	Bronze medal of the Physics University Olympiad Physics Olympiad for UNAM high schools.

JOURNAL PUBLICATIONS

- F. Huang and **V. Knapp-Perez**, “Cosmological stasis from field-dependent decay,” JHEP 09, 072 (2025) [doi:10.1007/JHEP09\(2025\)072](https://doi.org/10.1007/JHEP09(2025)072) [[arXiv:2502.20449](#) [[hep-ph](#)]].
- A. Baur, M. C. Chen, **V. Knapp-Perez** and S. Ramos-Sanchez, “Modular flavored dark matter,” JHEP 12, 091 (2024) [doi:10.1007/JHEP12\(2024\)091](https://doi.org/10.1007/JHEP12(2024)091) [[arXiv:2409.02178](#) [[hep-ph](#)]].
- **V. Knapp-Perez**, X. G. Liu, H. P. Nilles, S. Ramos-Sanchez and M. Ratz, “Matter matters in moduli fixing and modular flavor symmetries,” Phys. Lett. B 844, 138106 (2023) [doi:10.1016/j.physletb.2023.138106](https://doi.org/10.1016/j.physletb.2023.138106) [[arXiv:2304.14437](#) [[hep-th](#)]].
- M. C. Chen, **V. Knapp-Perez**, M. Ramos-Hamud, S. Ramos-Sanchez, M. Ratz and S. Shukla, “Quasi-eclectic modular flavor symmetries,” Phys. Lett. B 824, 136843 (2022) [doi:10.1016/j.physletb.2021.136843](https://doi.org/10.1016/j.physletb.2021.136843) [[arXiv:2108.02240](#) [[hep-ph](#)]].
- Y. Almumin, M. C. Chen, **V. Knapp-Pérez**, S. Ramos-Sánchez, M. Ratz and S. Shukla, “Metaplectic Flavor Symmetries from Magnetized Tori,” JHEP 05 (2021), 078 [doi:10.1007/JHEP05\(2021\)078](https://doi.org/10.1007/JHEP05(2021)078) [[arXiv:2102.11286](#) [[hep-th](#)]].
- A. Ayala, S. Hernandez-Ortiz, L. A. Hernandez, **V. Knapp-Perez** and R. Zamora, “Fluctuating temperature and baryon chemical potential in heavy-ion collisions and the position of the critical endpoint in the effective QCD phase diagram,” Phys. Rev. D 101 (2020) no.7, 074023 [doi:10.1103/PhysRevD.101.074023](https://doi.org/10.1103/PhysRevD.101.074023) [[arXiv:2003.04505](#) [[hep-ph](#)]].

PRE-PRINT PUBLICATIONS

- **V. Knapp-Perez**, G. Mohlabeng, M. Ratz and T. M. P. Tait, “Thermal History of Non-equilibrated Scalars,” [[arXiv:2507.21523](#) [[hep-ph](#)]].
- J. B. Baretz, M. Fieg, V. Ganesh, A. Ghosh, **V. Knapp-Perez**, J. Rudolph and D. Whiteson, “Towards AI-assisted Neutrino Flavor Theory Design,” [[arXiv:2506.08080](#) [[hep-ph](#)]].

REVIEWS

- Y. Almumin, M. C. Chen, M. Cheng, **V. Knapp-Perez**, Y. Li, A. Mondol, S. Ramos-Sanchez, M. Ratz and S. Shukla, “Neutrino Flavor Model Building and the Origins of Flavor and Violation,” Universe 9 (2023) no.12, 512 [doi:10.3390/universe9120512](https://doi.org/10.3390/universe9120512) [[arXiv:2204.08668](#) [[hep-ph](#)]], (Editor’s choice).

TEACHING AND MENTORING EXPERIENCE

OCT 2024-JUN 2025	Mentor for Undergraduate Research, University of California, Irvine (UCI) Supervised undergraduate research in particle physics; provided mentorship and guided study.
JUL 2024 & JUL 2025	Graduate Teaching Assistant at California State Summer School for Mathematics and Science (COSMOS), UCI Teaching assistant and mentor for high school students; delivered lectures and guided projects in particle physics and cosmology.
SEP 2021-MAR 2025	Graduate Teaching Assistant at UCI Taught lower-division courses (Classical Physics Laboratory, Basic Physics Laboratory, Modern Physics, Introduction to Mathematical Methods); upper-division courses (Classical Mechanics, Quantum Mechanics); and graduate courses (Electromagnetism, Quantum Mechanics).
FEB 2020-AUG 2021	Undergraduate Teaching Assistant at Science School, UNAM Taught undergraduate courses such as Differential Calculus, Differential Equations, Electromagnetism, Integral Calculus, and Quantum Mechanics.

PROFESSIONAL EXPERIENCE

2024	Referee for Journal of High Energy Physics (JHEP)
JUN 2023	Member, Local Organizing Committee, PASCOS 2023, UCI

TALKS

AUG 2025	“Autonomous Model Building Neutrino Flavor Theories with Reinforcement Learning”
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	Parallel talk at the Conferences on Supersymmetry and Unification of Fundamental Interactions (SUSY), University of California, Santa Cruz, USA.
MAY 2025	“Autonomous Model Building Neutrino Flavor Theories with Reinforcement Learning” Parallel talk at the Phenomenology 2025 Symposium, University of Pittsburgh, USA.
MAY 2023	“Metaplectic Flavor Symmetries from Magnetized Tori” Workshop talk at Mainz Institute for Theoretical Physics (MITP), Germany.
AUG 2023	“Stabilization of Modulus τ in Flavor Modular Symmetries” Seminar at Instituto de Física, UNAM, México.
JUN 2023	“Metaplectic Flavor Symmetries from Magnetized Tori” Parallel talk at the Particles, Strings, and Cosmology (PASCOS) Conference, UC Irvine, USA.
SEP 2022	“Modular Symmetries in Flavor Model Building” Seminar at Instituto de Física, UNAM, México.
MAR 2022	“Quasi-Eclectic Modular Symmetries” Seminar at String Pheno and BSM Physics Seminar, UNAM, México.
JUL 2021	“Metaplectic Flavor Symmetries from Magnetized Tori” Conference talk at the Division of Particles and Fields of the American Physical Society meeting (online).
APR 2021	“Flavor Symmetries from Donuts” Talk at the particle theory journal club at UCI, USA (online).
APR 2020	“The QCD Critical Point for Systems out of Equilibrium” Student Seminar at UNAM, Mexico.
DEC 2019	“Extended Isocurvature Constraints on QCD Axion Dark Matter” Student talk at ‘Estancias de Verano Teóricas’ 2020 contest, UNAM, Mexico.
MAY 2018	“Poincaré Group Representations: Spin $3/2$” Modern Physics Student Workshop at UNAM, Mexico.

SCHOOLS, CONFERENCES AND WORKSHOPS ATTENDED

AUG 2025	Conferences on Supersymmetry and Unification of Fundamental Interactions (SUSY) Conferences on research related to supersymmetric theories and other approaches to physics beyond the Standard Model in all aspects.
MAY 2025	2025 Phenomenology Symposium (PHENO) Symposium on the latest topics in particle phenomenology and theory as well as related issues in astrophysics and cosmology.
JUL 2024	74th Lindau Nobel Laureate Meeting International program bringing scientists together to attend lectures and engage with Nobel Laureates.
JUN 2024	Theoretical Advanced Study Institute in Particle Theory (TASI) Summer school featuring a pedagogical lecture series in theoretical particle physics.
MAY 2024	Modular Invariance Approach to the Lepton and Quark Flavour Problems: from Bottom-up to Top-down Workshop held at Mainz Institute for Theoretical Physics (MITP).
AUG 2023	51st SLAC Summer Institute Summer school in Machine Learning across the frontiers. Attended virtually.

JUN 2023	PASCOS 2023 Conference series on Particles, Cosmology, and Strings at UC, Irvine.
JUN 2021	IV 2021 Meeting of the Division of Particles and Fields of the American Physical Society Virtual meeting held at Florida State University (FSU).
OCT 2020	IV Mesoamerican Workshop on Cosmology and Gravity Workshop held at Nuclear Science Institute (ICN), UNAM.
OCT 2019	III Mesoamerican Workshop on Cosmology and Gravity Workshop held at Nuclear Science Institute (ICN), UNAM.
JUN 2019	ICTP Summer School on Particle Physics Summer school with lectures covering particle physics frontier topics.
JUL 2018	CINVESTAV Advanced Summer School Summer School aimed at undergraduate students, featuring frontier physics lectures from CINVESTAV and external researchers.
JUN 2018	UNAM Summer Physics School Summer School, aimed at undergraduate students, with frontier physics lectures delivered by UNAM researchers.

OUTREACH AND CONTRIBUTIONS TO DEI

SEP 2022-PRESENT	Mentor at PACE (Physics and Astronomy Community Excellence), UCI Provided one-on-one guidance and led workshops to support first-year graduate students' transition to graduate school.
SEP 2025	Panelist at Department of Physics & Astronomy, UCI (September 2025) Shared advice with incoming first-year graduate students on effective teaching practices.
FEB 2025	Volunteer at OC Discovery Cube, Science Museum Assisted with outreach and public engagement at the Solar System exhibit.
OCT 2024	Panelist at Society of Physics Students at UCI meeting Advised undergraduates on preparing for graduate school.
APR 2021	Panelist at Cal-Bridge Graduate School Workshop, UCI Provided guidance to students applying to Physics and Astronomy PhD programs.

SOFTWARE PACKAGES RELEASED

OCT 2025	FlavorBuilder Finite-group and model-building utilities interfacing with GAP
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COMPUTATIONAL SKILLS

- MATHEMATICA, LATEX, PYTHON, FORTRAN, JAVA, SLURM.
- MICROMEGAS, FEYNRULES.
- PANDAS, SYMPY, SCIKIT-LEARN, PYTORCH, OPENAI GYM.
- High-Performance Computing (HPC).

LANGUAGES

SPANISH: Native.
 ENGLISH: Fluent.
 ITALIAN: Basic proficiency.

PRESS AND MEDIA

2024	UCI Article on LEAD award https://ps.uci.edu/news/3081
2024	UC Article on UC President's Lindau Nobel Laureate Meetings Fellowship <a href="https://www.universityofcalifornia.edu/news/young-uc-scientists-selected-fello-
ws-prestigious-nobel-laureate-meeting">https://www.universityofcalifornia.edu/news/young-uc-scientists-selected-fello- ws-prestigious-nobel-laureate-meeting

CONTACT INFORMATION

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